

Research Plan: QIntern 2026

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1. Project Title

Benchmarking Efficient LCU Decompositions of Laplacian matrices for VQLS

2. Project Overview

The Variational Quantum Linear Solver (VQLS) [1] is a near-term hybrid quantum-classical algorithm for solving linear systems $A|x\rangle \propto |b\rangle$. Its efficiency depends critically on how the matrix A is decomposed as a linear combination of unitaries (LCU), $A = \sum_{l=1}^L c_l A_l$, since the number of distinct circuit evaluations required per VQLS iteration scales as $\mathcal{O}(L^2)$. For Laplacian matrices arising from finite-difference discretizations of PDEs, a naive Pauli decomposition yields $L = \mathcal{O}(N)$ terms which is exponential in the number of qubits ($n = \log_2 N$) making the standard formulation prohibitively expensive for large problem sizes.

A recent work [2] provides for efficient LCU decompositions for discrete d -dimensional Laplacians in arbitrary dimension with periodic, Dirichlet, Neumann, Robin, and mixed boundary conditions. These constructions use at most $10d$ terms with circuit depths and gate costs $\mathcal{O}(\log N)$. Separately, one of our works have explored block-encoding-based reformulations of VQLS to circumvent the decomposition cost altogether [3]. A systematic benchmarking study that evaluates VQLS scalability using efficient LCU decompositions to validate the need for the approach in [3] is lacking.

This project fills that gap. We build two end-to-end VQLS pipelines in Qiskit, both targeting Laplacian linear systems arising from 1D, 2D, and 3D finite-difference discretizations:

- **Standard VQLS:** The canonical formulation of [1] using Pauli LCU decomposition, with both local and global cost function variants. An existing implementation [4] (in an older Qiskit version) serves as the starting point; interns will modify this for the latest Qiskit version.
- **Efficient LCU VQLS:** VQLS integrated with the specialized, compact LCU decompositions for Laplacians developed in [2].

The two pipelines are then subjected to systematic benchmarks, with hardware runs on small instances. The central question is: *how much does replacing a generic Pauli decomposition with an efficient, structure-aware LCU reduce the practical circuit cost and iteration count of VQLS, and how does this advantage scale with problem dimension d , grid size N , and condition number κ ?*

3. Expected Background and Preparation

Interns are expected to have a solid understanding of basic quantum computing, linear algebra, and familiarity with variational quantum algorithms. A working knowledge of Python and prior experience with Qiskit is necessary. As preparation, interns are encouraged to read the VQLS paper [1], skim the LCU decomposition constructions in [2], and explore the LCU based vqls repository [4].

4. Work Plan (1 July – 15 August)

Week	Milestones
Week 1-2	Study the VQLS algorithm [1] and review the efficient LCU decompositions in [2]. Understand the structure of the existing VQLS codebase [4] and port this to the latest Qiskit version.
Week 3-4	Implement the efficient LCU decompositions from [2] in Qiskit for 1D, 2D, and 3D Laplacians with different boundary conditions. Validate circuit resources against analytical predictions. Integrate the efficient LCU circuits into the VQLS framework to assemble Pipeline 2.
Week 5-6	Conduct systematic benchmarks across both pipelines: vary N (grid size), d (dimension: 1D/2D/3D), boundary condition type, and condition number κ . Run small instances on IBM quantum hardware to assess the impact of noise on convergence. Draft the benchmarking paper and open-source release.

5. Deliverables

- An open-source Qiskit codebase containing both VQLS pipelines (standard Pauli LCU and efficient Laplacian LCU), with support for 1D/2D/3D problems and multiple boundary conditions.
- An arXiv paper presenting a systematic comparison of standard and efficient-LCU VQLS for Laplacian systems.

6. Guidance and Extension

I will be actively involved in both the theoretical and implementation aspects of the project, providing regular support via Discord to address queries. Weekly progress meetings will be held to review milestones, with additional meetings available upon request. A short introductory presentation on VQLS can be arranged at the start of the internship.

If bandwidth permits, several promising extensions can be explored. One direction is investigating whether the $\mathcal{O}(\log N)$ structure of the efficient LCU circuits motivates a correspondingly structured ansatz that avoids barren plateaus more effectively than hardware-efficient choices. Another direction is to explore preconditioners that reduces κ and thereby the required optimization iterations.

Intern-driven ideas for further development are highly encouraged.

References

- [1] Carlos Bravo-Prieto, Ryan LaRose, M. Cerezo, Yigit Subasi, Lukasz Cincio, and Patrick J. Coles. Variational quantum linear solver. *Quantum*, 7:1188, November 2023.
- [2] Thomas Hogancamp, Reuben Demirdjian, and Daniel Gunlycke. A linear combination of unitaries decomposition for the laplace operator, 2026.
- [3] Viraj Dsouza, Dhamotharan S, Tanishque Kumar, Ramesh Kolluru, Ferdin Don Bosco, Abhishek Chopra, Rut Lineswala, and Onkar Sahni. Towards hybrid quantum-classical computing for large-scale cfd: Tackling scalability challenges of the variational quantum linear solver. In *AIAA AVIATION FORUM AND ASCEND 2025*, pages AIAA 2025–3042, July 2025.
- [4] Viraj Dsouza and Rajas Dalvi. Tutorial_sci2025-hybrid_qls_for_cfd. https://github.com/virajd98/Tutorial_SCI2025-Hybrid_QLS_for_CFD, 2025. GitHub repository, accessed May 2026.